

Multiple Choice (10 marks)

1. Which of the following was not cited as evidence for life in the martian meteorite ALH 84001?
 - (a) Complex organic molecules specifically PAHs
 - (b) Magnetite grains similar to those formed by bacteria on Earth
 - (c) Carbonate minerals indicating a thicker warmer Martian atmosphere
 - (d) Amino acids with a preferred orientation or "chirality"
2. What Mars mission will be landing on May 25 2008 and will dig a trench into (hopefully) ice-rich soil?
 - (a) ExoMars
 - (b) Mars Exploration Rovers
 - (c) Mars Science Laboratory
 - (d) Phoenix Mars Lander
3. You observe the visible spectrum of a star viewed through a cloud of cool hydrogen gas.
 - (a) You see only the star's blackbody spectrum.
 - (b) You see the star's blackbody spectrum with absorption lines due to hydrogen.
 - (c) You see only emission lines characteristic of hydrogen.
 - (d) You see only emission lines characteristic of the star's composition.
4. Which of the following countries has sent landers to Venus?
 - (a) The U.S.S.R.
 - (b) The U.S.
 - (c) France
 - (d) A and B
5. By locating the north celestial pole (NCP) in the sky how can you determine your latitude?
 - (a) The azimuth of the NCP is the angular distance from the North Pole.
 - (b) The azimuth of the NCP is the same as your latitude.
 - (c) The altitude of the NCP is the same as your latitude.
 - (d) The altitude of the NCP is your angular distance from the North Pole.

True / False (10 marks)

Answer True or False

6. True or False: The Large Magellanic Cloud is a dwarf galaxy that is gravitationally bound to the Milky Way galaxy.
7. True or False: Type-Ia supernovae are primarily found in young galaxies where stellar formation is actively occurring.
8. True or False: If the Earth were moved to an orbit three times farther from the Sun, a year would be about 5.2 times longer due to Kepler's Third Law.
9. True or False: Sunspots typically appear in cycles of about 9 years, reflecting the Sun's magnetic activity variations.
10. True or False: The correct order of electromagnetic radiation from shortest to longest wavelength is gamma rays, X rays, ultraviolet, visible light, infrared, and radio.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the role of Jupiter's gravitational influence in preventing the formation of a planet within the asteroid belt, and analyze the implications for our understanding of planetary formation.
12. Discuss the reasons for the Mars Exploration Rover Spirit's northward tilt, particularly in relation to its location in the southern hemisphere during winter. Analyze the implications for its operations.

End of Paper